

T2-FLAIR Mismatch Sign Agreement Maximization Guide

A step-by-step procedure for aligning the automated *T2-FLAIR Mismatch Present* metric with radiologist annotations in the AKU-WHO dataset

Scope

This guide operationalizes the six remediation steps implemented in `btreport/additional_features/additional_features_3d_v2.py`. It explains the rationale for each step, the exact order in which to apply and verify them, how to read the Step-7 validation output, how to re-tune thresholds to maximize agreement, and how to resolve the failure modes that currently produce *Not Specified* results.

Context at a glance

Item	Value
Reference cohort	62 subjects annotated 'T2/FLAIR mismatch present' in the FLAIR Findings column
Cohort composition	39 Astrocytoma IDH-mutant, 17 Glioblastoma IDH-wildtype, 3 Astrocytoma IDH-mutant Grade 4, 3 Oligodendroglioma
Current pipeline agreement	0 agreements, 46 disagreements, 16 not specified (out of 62)
Dominant failure	Central FLAIR threshold (≤ 0.50) fails in 45 / 46 disagreement cases; median observed 0.78
Target after tuning	Sensitivity ≥ 0.80 on positives, specificity ≥ 0.70 on matched negatives

Companion script: `additional_features_3d_v2.py` | Validation command: `python -m ...additional_features_3d_v2 validate`

1. Why the current pipeline disagrees

Before changing anything, it is worth being precise about the failure. Comparing the *T2-FLAIR Mismatch Present* field of each subject's *_final.json against the radiologist annotation (all 62 reference subjects are annotated positive) produces zero agreements. The diagnostic breaks down as follows.

Condition in the binary rule	Pass rate (of 46 disagreements)	Median observed	Threshold
center FLAIR ≤ 0.50	1 / 46	0.777	≤ 0.50
score ≥ 0.12	31 / 46	0.184	≥ 0.12
center T2 ≥ 0.60	41 / 46	0.850	≥ 0.60
rim FLAIR ≥ 0.55	45 / 46	0.837	≥ 0.55

Table 1. The central-FLAIR check is the binding constraint: it fails in 45 of 46 disagreement cases. The binary rule requires all four conditions simultaneously.

Three structural causes

The numbers above are symptoms. The root causes are structural, and each maps to a remediation step.

Root cause	Evidence	Fixed by
The "center" is measured on the bright bulk of the tumor, not the FLAIR-suppressed cavity the radiologist annotates. The center/rim split uses $dt \geq 0.10 \cdot \max_dt$, which assigns roughly 90% of the lesion to "center" and a one-voxel shell to "rim".	Median central FLAIR is 0.78 on a [0,1] scale — that is bright tissue, not suppressed core.	Step 2
Normalization is whole-brain percentile-based, so the numeric value "0.50" means different things on different scanners and protocols.	EGD, BraTS and UCSF-PDGM subjects span multiple acquisition sources with no harmonization.	Step 3 + Step 4
The binary rule is a strict four-way AND, so any single noisy component forces a hard 0.	31 / 46 cases already pass the score gate but are still rejected by one component threshold.	Step 5
For a small set of subjects the BraTS segmenter labels the whole non-enhancing astrocytoma as edema, leaving no tumor core to score.	4 UCSF-PDGM subjects have Tumor Core Volume = 0.0 mL; EGD-0606, EGD-0523, EGD-0385 have TC/WT < 0.10.	Step 1 + Step 6

Table 2. Root causes and the remediation step that addresses each.

A note on the lesion mask

An early hypothesis was that the whole-tumor mask should replace the tumor-core mask everywhere. The data does not support that. In the 62 mismatch-positive subjects the median TC/WT ratio is 0.51, versus roughly 0.17–0.35 in non-mismatch baseline cases — the segmenter does find a substantial core for most mismatch lesions, and the literature describes the sign as a property of the tumor proper. The whole-tumor mask is therefore used only as a *fallback* for the seven subjects where the core genuinely collapses (Step 1), not as the default.

2. The six remediation steps

Each step is an independent boolean flag on the T2FlairMismatchV2 extractor. With every flag off the class reproduces the original behaviour exactly — this parity is what makes step-by-step verification meaningful: any change in the validation metrics is attributable to the one flag you turned on.

Step 1 — Tumor-core to whole-tumor fallback

What it changes. Keeps BraTS Tumor Core (NCR + ET) as the default lesion mask. Only when the core collapses — $TC < \text{min_core_volume_ml}$ (1.0 mL) or $TC/WT < \text{tc_wt_ratio_floor}$ (0.10) — does it fall back to Whole Tumor (NCR + ED + ET). The mask actually used is recorded in the `Lesion Mask Used` output field.

Why. Recovers the seven subjects where the segmenter assigned the entire non-enhancing astrocytoma to the edema label, while leaving the literature-aligned tumor-core default intact for the other 55.

Key parameters. `min_core_volume_ml = 1.0, tc_wt_ratio_floor = 0.10`

Step 2 — Center / rim geometry

What it changes. Replaces the $dt \geq 0.10 \cdot \text{max_dt}$ rule with a physical-distance split: the center is the deepest portion of the lesion (the deeper of `center_inward_mm = 3 mm` or 50% of the maximum inward depth), and the rim is the lesion minus the dilated center. Distances are computed in millimetres so the split is scanner-agnostic. A 95th-percentile-depth fallback guarantees a non-empty center for very small lesions.

Why. This is the highest-leverage fix. The original split sampled the bright tumor bulk as "center", which is exactly why central FLAIR read 0.78 instead of the suppressed value the radiologist sees. A small, genuinely deep center finally samples the FLAIR-suppressed cavity.

Key parameters. `center_inward_mm = 3.0, center_depth_fraction = 0.50, min_center_voxels = 10, min_rim_voxels = 10`

Step 3 — Contralateral white-matter normalization

What it changes. Divides the center / rim T2 and FLAIR means by the contralateral normal white-matter (CNWM) mean instead of applying whole-brain percentile min-max normalization. CNWM is derived from the SynthSeg merged segmentation (WM labels 2, 7 left; 41, 46 right); laterality selects the contralateral hemisphere.

Why. Puts every measurement on a subject-relative ratio scale. A center-FLAIR ratio near 1.0 means "suppressed to white-matter level" on any scanner; the absolute [0,1] value did not carry that meaning across the EGD / BraTS / UCSF-PDGM acquisition sources.

Key parameters. Requires `merged_seg`. Falls back gracefully under Step 6 if absent.

Step 4 — Ratio-scale thresholds

What it changes. Swaps the four absolute thresholds for ratio-scale values: center T2 ≥ 1.30 , center FLAIR ≤ 1.10 , rim FLAIR ≥ 1.30 , score ≥ 0.10 . These are the starting points; Section 5 explains how to re-tune them on the validation cohort.

Why. Absolute thresholds calibrated on percentile-normalized values are meaningless once Step 3 moves the measurements onto a ratio scale. Step 4 only behaves as documented when Step 3 is also on — the script emits a warning if it is enabled alone.

Key parameters. `ratio_center_t2_high = 1.30`, `ratio_center_flair_low = 1.10`, `ratio_rim_flair_high = 1.30`, `ratio_score_thresh = 0.10`

Step 5 — Soft weighted score

What it changes. Replaces the strict four-way AND with a soft score: each component (central T2 elevation, central FLAIR suppression, rim FLAIR elevation) is passed through a linear ramp to [0,1], the three are averaged, and a single threshold `soft_score_thresh` (0.40) decides the binary flag. The soft value is surfaced as *T2-FLAIR Mismatch Score*.

Why. The strict AND lets one noisy component veto an otherwise clear case — 31 / 46 disagreements already pass the score gate. A weighted average lets two strong components compensate for one weak one, which is closer to how a radiologist integrates the signs.

Key parameters. `soft_score_thresh = 0.40`, plus per-component ramp spans (`soft_*_span`, separate values for ratio vs. non-ratio mode)

Step 6 — Robust fallbacks

What it changes. Eliminates silent NaN returns from recoverable failures. If `merged_seg` is missing, it drops back to robust min-max normalization; if the CNWM mask is empty it uses bilateral WM, then the brain mask; if the center / rim is too small it retries with the whole-tumor mask. Every fallback is recorded in *Decision Reason*.

Why. 16 of 62 subjects currently return *Not Specified*. Most of those are recoverable structural failures, not genuine "cannot assess" cases. Step 6 converts them into scored results with an auditable reason string.

Key parameters. No tunable thresholds; behaviour is controlled by the other steps' parameters.

3. Operational runbook

Carry the steps out in the order below. The principle is to change one thing at a time and confirm the validation metrics move in the expected direction before enabling the next step.

3.0 Prerequisites

- Environment: `conda activate BTReport`; the script needs `numpy`, `scipy`, `pandas`, `nibabel` and `openpyxl`.
- Per-subject NIfTI inputs must be reachable. The harness auto-discovers `*_t2.nii.gz`, `*_flair.nii.gz`, `*_seg.nii.gz` (or `*_seg_pred.nii.gz`), `tmp/brain_mask.nii.gz` and `tmp/MNI152_in_subject_space_merged_seg.nii.gz` under each subject folder.
- The Excel annotation file (`aku_annotations_with_duplicates.xlsx`) and the `Dataset_AKU_WHO` root directory.

3.1 Establish the baseline (S0)

Run the harness with the baseline preset first. This must reproduce the original pipeline's result — 0 agreements, 46 disagreements, 16 not specified — on the 62 positives. If it does not, the file discovery or environment is wrong, and nothing downstream is trustworthy.

```
python -m btreport.additional_features.additional_features_3d_v2 validate \
--excel /path/to/aku_annotations_with_duplicates.xlsx \
--dataset /path/to/Dataset_AKU_WHO \
--presets S0 \
--n_negatives 62 \
--output results/v2_S0.csv
```

3.2 Enable steps cumulatively

Run the full cumulative ladder in a single command. The harness executes each preset in turn and prints one summary row per preset, so the marginal effect of every step is visible in one table.

```
python -m btreport.additional_features.additional_features_3d_v2 validate \
--excel /path/to/aku_annotations_with_duplicates.xlsx \
--dataset /path/to/Dataset_AKU_WHO \
--presets S0,S1,S12,S123,S1234,S12345,S123456 \
--n_negatives 62 \
--seed 0 \
--output results/v2_ladder.csv
```

This writes `results/v2_ladder.csv` (one row per preset) and `results/v2_ladder_per_subject.csv` (every subject's prediction, component values, lesion mask used and decision reason). The per-subject file is the primary input for threshold tuning in Section 5.

3.3 Expected trajectory

After preset	What should improve	What to check
S1	A few of the 16 Not Specified resolve (the TC-collapse subjects).	Lesion Mask Used = WT_fallback for ~7 subjects.
S12	Central FLAIR values drop sharply — median should move from ~0.78 toward ~0.3.	Central Core FLAIR Mean distribution in the per-subject CSV.

After preset	What should improve	What to check
S123	Measurements become ratio-scaled; raw thresholds no longer apply cleanly.	Expect this row to look noisy — that is why Step 4 follows immediately.
S1234	Sensitivity rises substantially as ratio thresholds bite.	sensitivity_excl_NS should clear ~0.5.
S12345	Sensitivity rises again; one weak component no longer vetoes a case.	sensitivity_excl_NS approaching the 0.80 target.
S123456	Not Specified count falls toward 0; specificity stabilizes.	NS_pos and NS_neg near 0.

Table 3. Qualitative trajectory. Exact numbers depend on the cohort and on the threshold tuning in Section 5; the point is the direction of travel.

3.4 Inspect individual subjects

When a single subject behaves unexpectedly, run it in isolation with verbose logging. The `single` subcommand prints the full output dict including the decision reason.

```
python -m btreport.additional_features.additional_features_3d_v2 single \
  --subject_folder /path/to/Dataset_AKU_WHO/Astrocytoma_IDH-mutant/BrATS2021_01494 \
  --steps S12345 --verbose
```

4. Reading the Step-7 validation output

The summary CSV has one row per preset. The columns split into raw counts and derived rates.

Column	Meaning
TP / FN	Positives predicted present / predicted absent.
NS_pos	Positives that returned NaN (Not Specified). Counted as a miss in the strict rate.
TN / FP	Negatives predicted absent / predicted present.
NS_neg	Negatives that returned NaN.
sensitivity_strict	$TP / (TP + FN + NS_pos)$. NaN treated as a miss — the honest number.
sensitivity_excl_NS	$TP / (TP + FN)$. NaN excluded — isolates the decision logic from coverage.
specificity_strict	$TN / (TN + FP + NS_neg)$.
specificity_excl_NS	$TN / (TN + FP)$.
agreement_pos / disagreement_pos / not_specified_pos	The three-way breakdown on the 62 positives, directly comparable to the original diagnostic.

Table 4. Summary CSV columns.

How to use the two sensitivity columns together

Read them as a pair. A large gap between `sensitivity_excl_NS` and `sensitivity_strict` means the decision logic is working but coverage is poor — the fix is Step 6 and the segmentation quality, not the thresholds. A low `sensitivity_excl_NS` with few NaNs means the thresholds themselves are wrong — go to Section 5.

The per-subject CSV is where tuning actually happens. Each row carries `Central Core T2 Mean`, `Central Core FLAIR Mean`, `Peripheral Rim FLAIR Mean`, `T2-FLAIR Mismatch Score`, `Lesion Mask Used` and `Decision Reason`. Group it by truth to see the separation between positives and negatives on each component.

5. Tuning methodology — maximizing agreement

Steps 1, 2, 3 and 6 are structural and are not tuned — they either work or expose a data problem. The agreement rate is maximized by tuning the *thresholds* introduced in Steps 4 and 5, in that order, against the validation cohort.

5.1 Tune the Step-4 ratio thresholds

With preset S123 producing ratio-scaled measurements, use the per-subject CSV to set each component threshold at the point of best separation between the 62 positives and the matched negatives.

- For each of the three components — `Central Core T2 Mean`, `Central Core FLAIR Mean`, `Peripheral Rim FLAIR Mean` — plot or tabulate the distribution split by truth.
- Pick the threshold at the crossover of the two distributions (equivalently, the value that maximizes Youden's $J = \text{sensitivity} + \text{specificity} - 1$ for that component alone).

- Set `ratio_center_t2_high`, `ratio_center_flair_low` and `ratio_rim_flair_high` to those crossover values.
- Re-run preset S1234 and confirm `sensitivity_excl_NS` improved without `specificity_excl_NS` collapsing.

5.2 Tune the Step-5 soft-score threshold

Once the component ramps are reasonable, `soft_score_thresh` is the single most important dial. Sweep it and pick the operating point.

- Run preset S12345 once, then in the per-subject CSV sweep `soft_score_thresh` from 0.25 to 0.60 in steps of 0.05, computing sensitivity and specificity at each value (the soft score is already in the CSV, so this is a spreadsheet operation — no re-run needed).
- Choose the lowest threshold that still meets the specificity floor (≥ 0.70). Lowering the threshold trades specificity for sensitivity.
- If sensitivity is still short of 0.80 at the specificity floor, widen the per-component ramp spans (`soft_*_span`) rather than dropping the threshold further — wider ramps make weak-but-present signs contribute more.

5.3 Order of adjustment when specificity collapses

If enabling a step or lowering a threshold causes specificity to fall below 0.70, adjust in this order — cheapest and most reversible first:

Priority	Adjustment	Effect
1	Raise <code>soft_score_thresh</code> .	Global precision/recall trade; one dial, no component re-tuning.
2	Tighten the single worst-separating component threshold.	Targets the component contributing the most false positives.
3	Narrow that component's ramp span.	Makes the soft score less forgiving of a borderline value.
4	Revisit Step 2 geometry (<code>center_depth_fraction</code>).	A deeper center sharpens the central-suppression signal but shrinks sample size.

Table 5. Escalation order for restoring specificity.

5.4 Acceptance criteria

Treat the tuning as done when, on preset S123456 with the tuned thresholds:

- `sensitivity_strict` ≥ 0.80 on the 62 positives;
- `specificity_excl_NS` ≥ 0.70 on the matched negatives;
- `NS_pos` + `NS_neg` is in single digits, and every remaining NaN has a `Decision Reason` that is a genuine "cannot assess" (empty tumor mask), not a recoverable failure.

Re-run with at least two different --seed values for the negative sample to confirm the specificity number is not an artefact of one particular draw.

5.5 Lock-in

- Write the tuned threshold values into the T2FLAIRMismatchV2 dataclass defaults, with a comment citing the validation run (date, cohort size, seed).

- Keep the per-subject CSV from the accepted run as the tuning record.
- Add a regression test that runs one mismatch-positive subject with all inputs present (for example BraTS2021_01494) under preset S123456 and asserts `T2-FLAIR Mismatch Present == 1`.
- Once S123456 is accepted, make it the default in the production pipeline by constructing the extractor with `StepFlags.from_string('all')`.

6. Parameter reference

All parameters are constructor arguments of `T2FlairMismatchV2`. Defaults reproduce the documented behaviour; the *Tune?* column flags the ones Section 5 adjusts.

Parameter	Default	Step	Tune?	Role
<code>min_core_volume_ml</code>	1.0	1	rarely	TC volume below which the WT fallback triggers.
<code>tc_wt_ratio_floor</code>	0.10	1	rarely	TC/WT ratio below which the WT fallback triggers.
<code>center_inward_mm</code>	3.0	2	rarely	Minimum inward depth (mm) for a voxel to count as center.
<code>center_depth_fraction</code>	0.50	2	yes	Center = deepest this fraction of the maximum inward depth.
<code>min_center_voxels</code>	10	2	no	Center smaller than this triggers the percentile fallback.
<code>min_rim_voxels</code>	10	2	no	Rim smaller than this returns NaN (or Step-6 retry).
<code>ratio_center_t2_high</code>	1.30	4	yes	Center T2 / CNWM must exceed this.
<code>ratio_center_flair_low</code>	1.10	4	yes	Center FLAIR / CNWM must be below this.
<code>ratio_rim_flair_high</code>	1.30	4	yes	Rim FLAIR / CNWM must exceed this.
<code>ratio_score_thresh</code>	0.10	4	yes	Legacy-style score gate on the ratio scale.
<code>soft_score_thresh</code>	0.40	5	yes	Soft-score decision threshold — the primary dial.
<code>soft_t2_span</code>	0.50	5	yes	Ramp width for the central-T2 component (ratio mode).
<code>soft_supp_span</code>	0.30	5	yes	Ramp width for the central-suppression component (ratio mode).
<code>soft_rim_span</code>	0.50	5	yes	Ramp width for the rim-FLAIR component (ratio mode).
<code>center_t2_high_thresh</code>	0.60	—	no	Original absolute threshold; used when Step 4 is off.
<code>center_flair_low_thresh</code>	0.50	—	no	Original absolute threshold; used when Step 4 is off.
<code>rim_flair_high_thresh</code>	0.55	—	no	Original absolute threshold; used when Step 4 is off.
<code>mismatch_score_thresh</code>	0.12	—	no	Original score gate; used when Step 4 is off.

Table 6. Constructor parameter reference for `T2FlairMismatchV2`. Non-ratio soft spans (`soft*_span_nonratio`) exist for the Step-5-without-Step-3 configuration and are not normally tuned.

7. Troubleshooting the failure modes

Every NaN result carries a `Decision Reason`. Match it against the table below.

Symptom / Decision Reason	Cause	Resolution
MISSING_FILES in the per-subject CSV	The harness could not find T2, FLAIR or segmentation for the subject.	Confirm the subject folder contains <code>*_t2.nii.gz</code> , <code>*_flair.nii.gz</code> and <code>*_seg.nii.gz</code> . Adjust naming to match, or extend the glob patterns in <code>discover_subject_files</code> .
NaN: lesion too small	Both TC and the WT fallback are below the voxel guard — a genuine empty / near-empty mask.	Inspect the segmentation. This is a legitimate "cannot assess" case; leave as NaN.

Symptom / Decision Reason	Cause	Resolution
NaN: center/rim too small	The lesion is too thin for the center/rim split even after the Step-6 WT retry.	Enable Step 6 if not already on. If it persists, the lesion is genuinely too small to assess.
NaN: step3 requires merged_seg	Step 3 is on but the SynthSeg merged segmentation is missing for the subject.	Run the anatomical-segmentation stage of the pipeline so <code>tmp/MNI152_in_subject_space_merged_seg.nii.gz</code> exists, or enable Step 6 to fall back to robust min-max normalization.
NaN: empty CNWM mask	The contralateral hemisphere WM labels are absent — usually a registration or laterality error.	Enable Step 6 (falls back to bilateral WM, then brain mask). Verify the laterality passed to the extractor; the harness infers it from the tumor centroid, which can be wrong for midline lesions.
Warning: Step 4 is on but Step 3 is off	Ratio thresholds are being applied to percentile-normalized [0,1] values.	Always enable Step 3 together with Step 4. Use the cumulative presets (S1234, not S124).
Specificity collapses when a step is enabled	The decision rule became too permissive.	Follow the escalation order in Table 5 — raise <code>soft_score_thresh</code> first.

Table 7. Failure-mode resolution.

Laterality note

The validation harness infers tumor laterality from the segmentation centroid. For lesions that cross or sit on the midline this guess can select the wrong contralateral hemisphere, which biases the CNWM reference. When a midline-crossing subject behaves oddly, pass an explicit laterality via the `laterality_overrides` argument of `validate()`, or via `--laterality` on the single subcommand.

End of guide. Companion script: `btreport/additional_features/additional_features_3d_v2.py`. Original implementation: `additional_features_3d.py`.